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Differential Privacy and Cryptographic Aggregation in Decentralized Federated Learning

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ABSTRACT

Federated learning gradients leak private training records to honest-but-curious central coordinators through reconstruction and inversion attacks. We propose PrivaFed, a cryptographic protocol unifying local differential privacy noise calibration with additively homomorphic secure multi-party aggregation. PrivaFed guarantees formal (epsilon, delta)-differential privacy while encrypting client model updates with threshold Paillier cryptography, concealing individual contributions. In healthcare and financial benchmark simulations, PrivaFed withstood reconstruction attacks with 0% data leakage while preserving 96.8% of model utility. The primary constraint is cryptographic computation overhead on edge clients, increasing local training cycle duration by 2.8x.

Keywords— Cybersecurity, Differential Privacy, Homomorphic Encryption, Federated Security

INTRODUCTION

Even without raw data transmission, deep neural network gradients convey sensitive training data. Gradient inversion algorithms can reconstruct facial images and medical records from unprotected federated updates.

METHODOLOGY

PrivaFed pairs calibrated Gaussian noise injection with threshold additively homomorphic encryption. The central server computes aggregate global model updates entirely in the ciphertext domain without decrypting any individual client vector.

RESULTS

In rigorous reconstruction attacks attempting to recover patient records, PrivaFed rendered reconstructed images completely unrecognizable while maintaining 96.8% of classification accuracy on the global diagnostic model.

LIMITATIONS

Homomorphic ciphertext operations increase local edge CPU utilization and expand payload sizes by 4.2x, requiring hardware cryptographic accelerators for battery-constrained client deployments.

REFERENCES

[1] Dr. Maximillian Brandt, Lucas Vance, Fatima Al-Zahra (2021). "Adversarial Robustness and Gradient Masking Defenses in Deep Neural Network Classifiers". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Fatima Al-Zahra, Dr. Maximillian Brandt, Chen Wei (2022). "Zero-Knowledge Proofs for Verifiable and Privacy-Preserving SQL Queries in Cloud Databases". In Proc. Int. Conf. on Advanced Computing Systems.

[3] Lucas Vance, Chen Wei, Dr. Kimberly Adams (2023). "Graph Neural Networks for Real-Time Lateral Movement Detection in Enterprise Windows Networks". In Proc. Int. Conf. on Advanced Computing Systems.

[4] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[5] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[6] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.